EdaphNet v2: Every File Explained

What each file is, what it uses, and what it feeds into · 3 October 2026 · repository github.com/Sudeep-Sharma1/edaphnet-v2

How to read this. Section 1 shows the whole project as one chain, from raw data to the paper. Section 2 traces one number from the paper back to the code, so you can see how the files connect. Sections 3 onwards go folder by folder. Each file has a coloured tag: Paper used for the paper's results · Record notes what was decided or found · Demo for showing the project · App the separate farmer app, not in the paper · Future for paper 2 (real photos) · Tool helper.

1. The whole project in one chain

Everything in the paper is produced by this chain. Each step reads the output of the step before it.

1 Data `download_ossl.py` gets 3 OSSL files into `data/raw/`	**2 Clean datasets** `build_ossl_dataset.py` makes `data/processed/t1_*.npz` and splits	**3 Witness** `fit_witness_ladder.py` makes colour/spectrum predictions `witness_*.npz`
4 Experiments `configs/*.yaml` + `run_real.py` use the method in `src/edaphnet2/`	**5 Results** run folders in `results/runs/`; analysis scripts write `results/exploratory/`	**6 Writing** docs 11–26, figures, `docs/paper/draft_v0.md`, PDFs in `reports/`

1 → 2 → 3 → 4 → 5 → 6 (read left to right, top row then bottom row)

Three side branches use pieces of the chain but do not feed the paper:

- **Demos:** `train_colour_brain.py` builds a colour model in `models/`, and `serve_demo.py` uses it with the method to run `demo_site/live.html`. `build_demo_page.py` builds the offline page `demo_site/index.html` from the run records.
- **Future (paper 2):** `phone.py` + `process_photos.py` turn real soil photos into witness values, following the protocol in doc 15 and the sheets in `templates/`.
- **App:** everything under `app/` plus the `app_*.py` scripts and `data/app_raw/`.

2. Follow one number from the paper to the code

Example: the paper says **"with a spectrometer, EdaphNet found +0.092 more errors than the best existing method"**.

1. **Paper:** `docs/paper/draft_v0.md` §7.4, table row "Spectrometer".
2. **Document that reports it:** `docs/21_kssl_preregistered_results.md`, the pre-registered verdict.
3. **Rules it was judged by:** `docs/18_preregistered_amendment_v2.md`, written *before* the test.
4. **Script that computed the verdict:** `scripts/analyse_amendment.py`.
5. **Run folder it read:** `results/runs/20261001-234902_real_kssl3_095deaa/` (`amendment_A1.csv`, `metrics_by_setting.csv`).
6. **Settings used:** `configs/real_kssl3.yaml` (copied into the run folder as `config.json`).
7. **Program that ran it:** `scripts/run_real.py`, which calls the method in `src/edaphnet2/inference.py` and the baselines in `src/edaphnet2/baselines.py`.

8. **Data it used:** `data/processed/t1_kssl3.npz` and `witness_kssl3.npz`, built from `data/raw/ossl/`.
9. **Proof of the code version:** `provenance.json` in the run folder (commit 095deaa).

Every other result works the same way. `REPRODUCE.md` lists the command behind each one.

3. Top-level files

File	What it is (simple words)	Connected to
`README.md` Record	The front page: the problem, progress, findings, mistakes corrected, folder guide, data licences.	Points to every other guide
`HANDOVER.md` Record	Your to-do list after 5 October: guide meeting, literature search, citations, IP office, Zenodo, submission.	Faculty report, paper
`REPRODUCE.md` Paper	The exact commands from data download to every table and figure.	All scripts and run folders
`HOW_TO_RUN.md` Tool	How to install, test, run demos and experiments; troubleshooting.	Scripts, demos
`FILE_GUIDE.md` Record	Short table of all files (this PDF is the long version).	—
`requirements.txt`, `pytest.ini`, `pyrefly.toml` Tool	Python packages; test settings; editor type-checker settings.	tests, editor
`.gitignore`, `.gitattributes` Tool	Keep big/private files off GitHub (raw data, models, app images, `invention_log.md`); line-ending rules.	git

4. `src/edaphnet2/`: the method (the heart of the project)

This is the research code. It was **frozen at commit 095deaa** before the pre-registered test, so the test could not be tuned afterwards.

File	What it does	Uses → / used by
`inference.py` Paper	**The method.** Looks at a whole batch of 24 reports at once, tries every likely swap and every error type, and gives each report a suspicion score, a most likely explanation and a suggested correction.	uses `operators`, `swaps`, `witness`, `space` · used by `run_real.py`, `run_sim.py`, demos
`operators.py` Paper	The list of value errors: ×10, ÷10, unit mix-up, swapped digits, swapped columns, gross error. Also plants random errors for testing.	used by `inference`, `simulate`, `run_real`
`swaps.py` Paper	The swap patterns: neighbours swapped, a whole tray shifted by one or two places, any two samples swapped.	used by `inference`
`witness.py` Paper	Witness models: from colour or spectrum, predict each sample's carbon, nitrogen etc. with an uncertainty.	used by `inference`, `fit_witness_ladder`, `serve_demo`
`baselines.py` Paper	The 7 competitors B0–B7 (range checks, soil rules, lab-only model, witness interval, swap matcher, supervised classifier, cleanlab) and B6 rule-based repair.	used by `run_real`, `run_sim`
`conformal.py` Paper	False-alarm control: turns scores into p-values using clean calibration reports, then Benjamini–Hochberg.	used by `run_real`, `rescore_pools`

`metrics.py` Paper	How results are measured: errors found when re-testing the top 5%, power, false-alarm rate, per pool of 50 batches.	used by `run_real`, `run_sim`
`space.py` Paper	How each property is stored and printed (decimal places, log scale), so planted errors look like real printed reports.	used by almost every module
`real_data.py` Paper	Loads the soil data; defines the property sets (ICRAF 8, KSSL 3); computes soil colour from a spectrum.	used by `build_ossl_dataset`, `run_real`
`properties.py`, `simulate.py` Paper	Make the synthetic soils, batches and errors for the first test (simulation).	used by `run_sim`
`tracking.py` Paper	Writes the run record: settings, code version, computer, timing. This is what makes every number traceable.	used by every experiment
`phone.py` Future Demo	Real-photo processing: colour-card correction, true soil colour, blur and exposure checks, texture.	used by `process_photos`, `serve_demo`

5. `scripts/`: programs you run

Data and experiments Paper

File	What it does	Reads → writes
`download_ossl.py`	Downloads the 3 soil data files and checks their fingerprints (MD5).	internet → `data/raw/ossl/`
`ossl_audit.py`	Checks the data: what is usable, licences, how many samples per property (doc 12).	raw data → `results/data_audit/`
`build_ossl_dataset.py`	Makes the clean ICRAF and KSSL datasets, split by location so test soils come from places never seen in training.	raw → `data/processed/t1_*.npz`, `data/splits/`
`fit_witness_ladder.py`	Trains the colour / bands / spectrum witnesses and measures how much each one knows (R^2).	processed → `witness_*.npz`, run folder
`run_sim.py`	Runs the simulation experiment (doc 13).	`configs/sim_t0.yaml` → run folder
`run_real.py`	Runs the real-data experiments: plants errors in real reports, runs the method and all baselines, saves every score. Can resume if interrupted.	config + data → run folder
`mismatch_ops.py`	Error types the method was *not* told about (×100, big gross, copied neighbour, dropped digit), for the "unknown errors" test.	used by `run_real.py`
`diag_calibration.py`	Found the calibration mistake (the same samples counted many times).	data → run folder

Analyses Paper

File	What it answers	Reads → writes
`analyse_amendment.py`	Did the pre-registered claims pass or fail? (doc 21)	KSSL run → `amendment_*.csv`
`summarise_runs.py`	Summary tables of any run.	run folder → tables
`detectability_table.py`	Which error types can each witness find? (doc 19)	run → table

`exploratory_tables.py`	Swaps vs wrong values; what each part of the method adds (doc 23, doc 26).	run → `results/exploratory/`
`rescore_pools.py`	How false-alarm control depends on how many reports are tested at once (Figure 5).	run → `rescore_pools.csv`
`diag_gross.py`	Why gross errors are rarely found (doc 26 §1).	run → `gross_*.csv`
`error_mix.py`	Who wins as the share of swaps changes; draws Figure 6 (doc 26 §2).	run → `error_mix.csv`, figure
`make_paper_figures.py`	Draws paper Figures 1–5 from the run records (no numbers typed by hand).	runs → `docs/paper/figures/`
`build_report_pdf.py`	Turns the paper or progress report into a PDF.	markdown → `reports/*.pdf`

Demos, future and app

File	What it does	Connected to
`demo.py` Demo	Command-line demo: one batch with a planted swap and ×10 error, and the re-test list.	method, KSSL data
`serve_demo.py` Demo	Small web server for the live demo page (photo + report → result).	`demo_site/live.html`, `models/`, method
`train_colour_brain.py` Demo	Trains the colour → carbon/nitrogen model the live demo uses.	→ `models/colour_brain_*.joblib`
`build_demo_page.py` Demo	Builds the offline presentation page with real demo batches and result charts.	run records → `demo_site/index.html`, `web.html`
`process_photos.py` Future	Turns a folder of real soil photos into witness values (doc 15).	`phone.py`, `templates/`
`app_*.py` (5 files) App	Build the crop list, give crop advice, audit leaf images, extract leaf features, train the leaf model.	`app/`, `data/app_raw/`

6. `configs/` and `tests/`

Config Paper	Experiment
`sim_t0.yaml`	Simulation
`real_icraf8.yaml`	First real test (ICRAF)
`real_kssl3.yaml`	**Pre-registered KSSL test**
`real_kssl3_mismatch.yaml`	Unknown errors, 1 seed
`real_kssl3_mismatch5.yaml`	Unknown errors, 5 seeds

Test file (34 tests)	Checks
`test_core.py`	Method, errors, swaps, false-alarm control
`test_real_data.py`	Data loading, colour from spectra
`test_resume.py`	Resumed run = same numbers (slow)
`test_phone.py`, `test_process_photos.py`	Photo processing
`test_app_*.py`	App crops, disease sources, server

7. `docs/` : the written record, in the order things happened

Document	What it says	Leads to
`08` research and patent charter Record	Goals and integrity rules: no invented results, rules fixed before tests, failures reported.	everything
`09`, `10` prior art Record	What already exists (16 papers/patents/products compared) and what seemed new.	doc 11, paper §2
`11` problem specification Paper	The plan: model, error types, baselines, hypotheses H1–H6 and pass/fail rules (pre-registration v1).	docs 13, 17
`12` data audit Paper	Checked the soil data; includes a corrected counting mistake.	datasets
`13` simulation results Paper	The method works only when the witness is moderately informative; gains come from swaps.	paper §7.1
`14` witness ladder Paper	How much colour knows about real soils (mainly carbon).	paper §7.2
`15` step-6 protocol Future	How to collect real photos + lab reports with blind duplicates.	paper 2
`16` paper skeleton Record	Early outline of the paper.	draft_v0
`17` first real test (ICRAF) Paper	Failed; later traced to our calibration mistake (correction note inside).	docs 18, D-016
`18` amendment v2 Paper	New pre-registered rules for the KSSL test (A1–A4, 5% re-test budget).	doc 21
`19` detectability Paper	Which error types each witness can find.	paper §7.5
`20` amendment v3 draft Future	Plan for the next test: a combined re-test policy, LUCAS confirmation.	next study
`21` KSSL verdict Paper	**The main result:** A1 failed; A1-c, A2, A3 passed.	paper §7.4
`22` progress report Record	Plain-language status on 2 October (source of the progress PDF).	reports/
`23` exploratory Paper	Why A1 failed: swaps vs values, ablations, pool size, unknown errors.	paper §7.5–7.7
`24`, `25` app design and data App	The farmer app's design and its data audit.	app/
`26` review fixes Paper	After the external review: gross errors, error mix, 5 seeds, text fixes.	paper v0.2
`decision_log.md` Record	Every decision D-001 to D-025, dated, with the reason.	all docs
`search_log.csv` Record	Literature-search log (the formal Scopus/WoS search must still be added).	paper §2
`paper/draft_v0.md` + `figures/` Paper	**The paper** and its 6 figures (file numbers are one lower than figure numbers).	reports/ PDF
`invention_log.md` Record	Invention notes; on your laptop only, never on GitHub.	IP office

8. `results/`, `data/`, `models/`

Path	What is in it	On GitHub?

`results/runs/<date>_<name>_<commit>/` Paper	One folder per experiment: `config.json` (settings), `provenance.json` (code version), `metrics_by_setting.csv`, `attribution_by_setting.csv`, `predictions.csv.gz` (every report's score; laptop only).	yes, except predictions
`results/exploratory/<run>/` Paper	Tables from the analysis scripts.	yes
`results/data_audit/` Paper	Data audit and dataset build summaries.	yes
`results/*.log`, `results/logs/` Record	Screen output of the long runs.	yes
`results/app_audit/`, `colour_brain_report.json`	App image audits; demo colour-model accuracy.	yes
`data/ossl_manifest_md5.txt`, `data/splits/` Paper	Fingerprints of the 3 data files; which sample is in train / calibration / test.	yes
`data/raw/`, `data/processed/` Paper	The soil data files and clean datasets (≈ 200 MB; rebuild with `download_ossl.py`; rebuild checked identical).	no
`data/app_raw/` App	PlantVillage, PlantDoc, EcoCrop and leaf features.	no
`models/` Demo	Demo colour models (and the leaf model once trained).	no

9. `reports/`, `demo_site/`, `templates/`, `app/`

File	What it is
`reports/EdaphNet_Paper_Draft_v0.pdf`	The paper (v0.2) as PDF, with all 6 figures.
`reports/EdaphNet_Faculty_Discussion.pdf`	Status, problems and decisions for your guide (written before the review fixes).
`reports/EdaphNet_Progress_Report.pdf`	Earlier progress report (2 October).
`reports/EdaphNet_Project_Summary.pdf`	The complete project summary (companion to this PDF).
`demo_site/index.html`, `web.html` Demo	Offline presentation page (double-click); `web.html` is the version for the web link; `template.html` is its template.
`demo_site/live.html` Demo	Live demo with animations and Guided demo; needs `serve_demo.py` running.
`templates/*.csv` Future	Sheets for collecting samples, photos, lab results and the secret blind-duplicate key.
`app/server.py`, `app/web/index.html` App	App backend and 4-tab phone page (leaf, soil, crops, report check).
`app/crop/`, `app/leaf/`, `app/knowledge/` App	Crop scoring (FAO EcoCrop rules), leaf model, sourced crop and disease information.

10. Words you will see

Word	Meaning

Witness	A cheap, independent look at the same soil sample (its colour, or a spectrum) that the lab cannot corrupt.
Swap	A report that belongs to a different sample (labels or tray positions mixed up).
Value error	One wrong number in a report (×10 slip, unit mix-up, swapped digits, gross error).
Batch / pool	A batch is 24 reports (one lab tray); a pool is 50 batches (about one lab-day).
Re-test budget	The share of reports the lab can afford to re-test; we use the top 5% most suspicious.
Baseline (B0–B7)	An existing or simple method we compare against; B5 (supervised) was usually the strongest.
Pre-registration	Writing the pass/fail rules down and committing them *before* running the test.
Exploratory	An analysis done after seeing results; useful for ideas, not proof.
FDR / false alarms	The share of flagged reports that were actually fine; we aim for 10%.
Calibration	Clean reports used to decide how unusual a score is; reusing the same samples was our mistake.
Run folder / provenance	The saved record of one experiment, including the exact code version.